YardWise | API Integration Guide

API comparison

Service	Purpose	How YardWise uses it	Free/public limitations
PlantSolve	Indoor plant care	Houseplant profiles, care and watering; import only if license permits	Advertises no rate caps; attribution; verify terms
OpenPlantDB	Outdoor gardening	Download CC0 data; build search, planting windows and zone filters	No published hosted cap; self-host dataset
USDA PLANTS	Native plants, trees, shrubs	Import relevant species and distribution data	Public data; verify individual images/asset rights
GBIF	Plant taxonomy worldwide	Validate scientific names and enrich species index	Free with throttling; check source licenses
NOAA/NWS	US weather and alerts	Backend forecast lookup, frost/rain alerts, weather-aware tasks	Free; fair-use throttling; US only
Open-Meteo	Global weather	Alternative worldwide forecasts	Free hosted tier noncommercial with quotas; commercial plan needed
USDA Hardiness	US growing zones	Map user location to hardiness zone for suitability	Public map/data; check redistribution rules
Nominatim/OSM	Geocoding	Convert addresses/ZIP codes to coordinates; cache results	Public service has strict rate and usage policy
openrouteservice	Professional routing	Estimate travel for crew/client visits	Free tier with usage limits
Supabase	Application data	Auth, gardens, projects, budgets, plant records and cached forecasts	Free tier quotas; RLS required

React + JavaScript, Express + Node.js, Supabase

Public website for homeowners and landscaping professionals

Suggested JavaScript API routes

- **/api/plants —** Search cached indoor/outdoor catalog
- **/api/plants/:id —** Plant detail from Supabase
- **/api/plants/recommend —** Match climate, sunlight, space and goals
- **/api/weather —** Cached local forecast
- **/api/weather/alerts —** Weather warnings and task recommendations
- **/api/estimates —** Calculate soil, mulch, pavers and other materials
- **/api/projects/:id/layout —** Saved landscape design
- **/api/maintenance —** Personalized care tasks
- **/api/clients —** Private professional client management

Recommended implementation order

1. 1. Import OpenPlantDB data and build a searchable catalog in Supabase.
2. 2. Add indoor plant care data from PlantSolve.
3. 3. Integrate NOAA/NWS for US weather and alerts; cache forecasts by area and time.
4. 4. Add USDA hardiness and USDA PLANTS for outdoor suitability and native plants.

5. 5. Use GBIF for name validation and additional taxonomy, not watering instructions.

Documentation links

- PlantSolve: https://www.plantsolve.com/developers/
- OpenPlantDB: https://plants.fairbrook.org/api
- USDA PLANTS: https://plants.sc.egov.usda.gov/downloads
- GBIF: https://www.gbif.org/developer/summary
- NOAA/NWS: https://www.weather.gov/documentation/services-web-api
- Open-Meteo: https://open-meteo.com/en/docs
- USDA Hardiness: https://planthardiness.ars.usda.gov/
- Nominatim/OSM: https://nominatim.org/release-docs/latest/api/Overview/
- openrouteservice: https://openrouteservice.org/dev/#/api-docs
- Supabase: https://supabase.com/docs